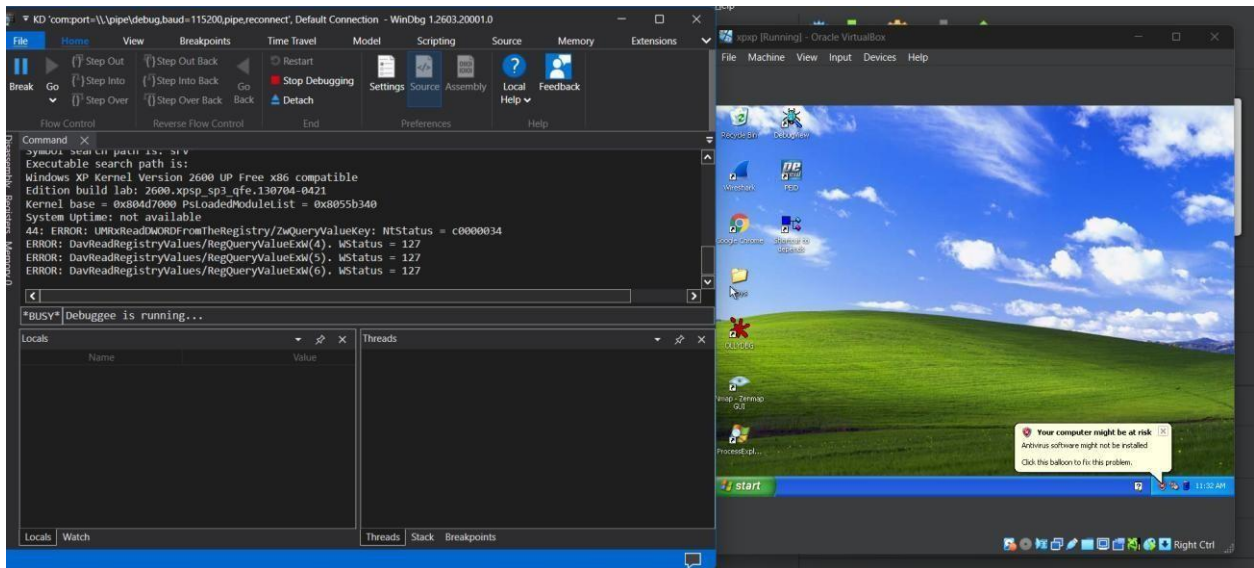


The final solo lab of the semester for CSC471 revolves around the rootkit, a type of malicious software that impacts the system at the kernel level. The below screenshot confirms the successful installation of WinDBG onto my host machine (HP ENVY laptop), as well as a successful run of the windows XP environment in Virtual Box. This screenshot also confirms the two are connected through the ports as instructed.



The SSDT is the “System Service Descriptor Table” essentially, a translation table that provides the memory address of each system call. According to the lecture slides this table is associated with “ring 0”, or the privileged kernel layer of the ring. It resides in read-only memory but, can be used as a location for a malicious user to hook into.

The first column is the location for a table entry. This index is essentially where the syscall pointer is on the table. The second column is for the actual function pointer. The third column is the symbol or human-readable name for the syscall. Below is the full output of the table before the executable is ran.

```
Command x
kld -dds KiServiceTable L 12a
804e2620 8057a521 nt!NTAcceptConnectPort
804e2624 805760c9 nt!NTAccessCheck
804e2628 8058cabf nt!NTAccessCheckAndAuditAlarm
804e262c 8058e3f7 nt!NTAccessCheckByType
804e2630 8058d7c0 nt!NTAccessCheckByTypeAndAuditAlarm
804e2634 80638f02 nt!NTAccessCheckByTypeResultList
804e2638 8063b113 nt!NTAccessCheckByTypeResultListAndAuditAlarm
804e263c 8063b15c nt!NTAccessCheckByTypeResultListAndAuditAlarmByHandle
804e2640 805745ef nt!NTAddAtom
804e2644 80649b8b nt!NTSetBootEntryOrder
804e2648 8063873d nt!NTAdjustGroupsToken
804e264c 8058c0de nt!NTAdjustPrivilegeToken
804e2650 80630120 nt!NTAlertResumeThread
804e2654 80577310 nt!NTAlertThread
804e2658 80591595 nt!NTAllocateLocallyUniqueId
804e265c 8062702f nt!NTAllocateUserPhysicalPages
804e2660 8059df59 nt!NTAllocateUids
804e2664 80569302 nt!NTAllocateVirtualMemory
804e2668 805d7867 nt!NTAreMappedFilesTheSame
804e266c 805a1387 nt!NTAssignProcessToJobObject
804e2670 804e2c2c nt!NTCallbackReturn
804e2674 80649b77 nt!NTDeleteBootEntry
804e2678 805c99ad nt!NTCancelIoFile
804e267c 804ec61c nt!NTCancelTimer
804e2680 805699ae nt!NTClearEvent
804e2684 80567c07 nt!NTClose
804e2688 8058d24c nt!NTCloseObjectAuditAlarm
804e268c 8064f9c0 nt!NTCompactKeys
804e2690 8058b776 nt!NTCompareTokens
804e2694 8057ac9b nt!NTCompleteConnectPort
804e2698 8064fc2d nt!NTCompressKey
804e269c 8058cb11 nt!NTConnectPort
```

```
Command X
804e26a0 804e1f6a nt!NTContinue
804e26a4 8065b0f6 nt!NTCreateDebugObject
804e26a8 805a174f nt!NTCreateDirectoryObject
804e26ac 8057013a nt!NTCreateEvent
804e26b0 8064a1dc nt!NTCreateEventPair
804e26b4 8056f980 nt!NTCreateFile
804e26b8 8058e635 nt!NTCreateIoCompletion
804e26bc 805aa070 nt!NTCreateJobObject
804e26c0 806305cb nt!NTCreateJobSet
804e26c4 80573887 nt!NTCreateKey
804e26c8 805d7758 nt!NTCreateMailslotFile
804e26cc 805776e0 nt!NTCreateMutant
804e26d0 805874b1 nt!NTCreateNamedPipeFile
804e26d4 805c2c0b nt!NTCreatePagingFile
804e26d8 805dc20a nt!NTCreatePort
804e26dc 805b02e9 nt!NTCreateProcess
804e26e0 805830e4 nt!NTCreateProcessEx
804e26e4 8064a7fd nt!NTCreateProfile
804e26e8 80565433 nt!NTCreateSection
804e26ec 8057d6a5 nt!NTCreateSemaphore
804e26f0 8059e796 nt!NTCreateSymbolicLinkObject
804e26f4 80578925 nt!NTCreateThread
804e26f8 805dcc22 nt!NTCreateTimer
804e26fc 805a7a24 nt!NTCreateToken
804e2700 805d9a06 nt!NTCreateWaitablePort
804e2704 8055c271 nt!NTDebugActiveProcess
804e2708 8065c3cb nt!NTDebugContinue
804e270c 80566590 nt!NTDelayExecution
804e2710 8058bef5 nt!NTDeleteAtom
804e2714 80649b77 nt!NTDeleteBootEntry
804e2718 805d610b nt!NTDeleteFile
804e271c 80595aba nt!NTDeleteKey
804e2720 8063b1b7 nt!NTDeleteObjectAuditAlarm
804e2724 805c3a2f nt!NTDeleteObjectAuditAlarm
```

```
Command X
804e2724 805936da nt!NTDeleteValueKey
804e2728 805796db nt!NTDeviceIoControlFile
804e272c 805c5e6b nt!NTDisplayString
804e2730 805749da nt!NTDuplicateObject
804e2734 8058081b nt!NTDuplicateToken
804e2738 80649b8b nt!NTSetBootEntryOrder
804e273c 80573f95 nt!NTEnumerateKey
804e2740 80649b63 nt!NTEnumerateSystemEnvironmentValuesEx
804e2744 805819e3 nt!NTEnumerateValueKey
804e2748 80625e50 nt!NTExtendSection
804e274c 805afad0 nt!NTFilterToken
804e2750 805961e0 nt!NTFindAtom
804e2754 8058c894 nt!NTFlushBuffersFile
804e2758 8056e7aa nt!NTFlushInstructionCache
804e275c 8059d120 nt!NTFlushKey
804e2760 805960e1 nt!NTFlushVirtualMemory
804e2764 80627893 nt!NTFlushWriteBuffer
804e2768 806273e4 nt!NTFreeUserPhysicalPages
804e276c 80569c2d nt!NTFreeVirtualMemory
804e2770 805771f8 nt!NTFsControlFile
804e2774 805e056b nt!NTGetContextThread
804e2778 8062c893 nt!NTGetDevicePowerState
804e277c 8059f066 nt!NTGetPlugPlayEvent
804e2780 8053b86d nt!NTGetWriteWatch
804e2784 805dc22e nt!NTImpersonateAnonymousToken
804e2788 8058b1e2 nt!NTImpersonateClientOfPort
804e278c 805817c1 nt!NTImpersonateThread
804e2790 805a6f30 nt!NTInitializeRegistry
804e2794 8062c05f nt!NTInitiatePowerAction
804e2798 8063047f nt!NTIsProcessInJob
804e279c 8062c87a nt!NTIsSystemResumeAutomatic
804e27a0 805a95b1 nt!NTListenPort
804e27a4 805a29bd nt!NTLoadDriver
```

```
Command X
804e27a8 805adcb0 nt!NTLoadKey
804e27ac 805adaf8 nt!NTLoadKey2
804e27b0 8059095c nt!NTLockFile
804e27b4 805afce0 nt!NTLockProductActivationKeys
804e27b8 805d0d8f nt!NTLockRegistryKey
804e27bc 805af122 nt!NTLockVirtualMemory
804e27c0 8059e0e3 nt!NTMakePermanentObject
804e27c4 8059eb60 nt!NTMakeTemporaryObject
804e27c8 8062651b nt!NTMapUserPhysicalPages
804e27cc 806269ef nt!NTMapUserPhysicalPagesScatter
804e27d0 8057cb01 nt!NTMapViewOfSection
804e27d4 80649b77 nt!NTDeleteBootEntry
804e27d8 80591d22 nt!NTNotifyChangeDirectoryFile
804e27dc 80591a6b nt!NTNotifyChangeKey
804e27e0 80591b34 nt!NTNotifyChangeMultipleKeys
804e27e4 8057b28f nt!NTOpenDirectoryObject
804e27e8 80581b30 nt!NTOpenEvent
804e27ec 8064a2cd nt!NTOpenEventPair
804e27f0 8056f91b nt!NTOpenFile
804e27f4 80616e9f nt!NTOpenIoCompletion
804e27f8 80630823 nt!NTOpenJobObject
804e27fc 80569080 nt!NTOpenKey
804e2800 8057778e nt!NTOpenMutant
804e2804 80592ba5 nt!NTOpenObjectAuditAlarm
804e2808 80574bc1 nt!NTOpenProcess
804e280c 80571121 nt!NTOpenProcessToken
804e2810 8057131a nt!NTOpenProcessTokenEx
804e2814 8056e583 nt!NTOpenSection
804e2818 805daca3 nt!NTOpenSemaphore
804e281c 8057b15b nt!NTOpenSymbolicLinkObject
804e2820 80598c4c nt!NTOpenThread
804e2824 80570bbe nt!NTOpenThreadToken
804e2828 80570b2f nt!NTOpenThreadTokenEx
804e282c 80570b2f nt!NTOpenThreadTokenEx
```

Command	
804e282c	8064a103 nt!NtOpenTimer
804e2830	8059be7a nt!NtPlugPlayControl
804e2834	8059abdc nt!NtPowerInformation
804e2838	80596c0c nt!NtPrivilegeCheck
804e283c	8059dde8 nt!NtPrivilegeObjectAuditAlarm
804e2840	805a96f4 nt!NtPrivilegedServiceAuditAlarm
804e2844	80574f70 nt!NtProtectVirtualMemory
804e2848	805d9a2e nt!NtPulseEvent
804e284c	805707ea nt!NtQueryAttributesFile
804e2850	80649b8b nt!NtSetBootEntryOrder
804e2854	80649b8b nt!NtSetBootEntryOrder
804e2858	804f97e5 nt!NtQueryDebugFilterState
804e285c	80566d02 nt!NtQueryDefaultLocale
804e2860	80581f1f nt!NtQueryDefaultUILanguage
804e2864	8057d37b nt!NtQueryDirectoryFile
804e2868	80587913 nt!NtQueryDirectoryObject
804e286c	806170e8 nt!NtQueryEaFile
804e2870	8057b38c nt!NtQueryEvent
804e2874	8058084a nt!NtQueryFullAttributesFile
804e2878	805d57e8 nt!NtQueryInformationAtom
804e287c	8057bc38 nt!NtQueryInformationFile
804e2880	80583d25 nt!NtQueryInformationJobObject
804e2884	80623917 nt!NtQueryInformationPort
804e2888	80570d5c nt!NtQueryInformationProcess
804e288c	8056731e nt!NtQueryInformationThread
804e2890	8057188b nt!NtQueryInformationToken
804e2894	80581c74 nt!NtQueryInstallUILanguage
804e2898	8064acaf nt!NtQueryIntervalProfile
804e289c	80616f60 nt!NtQueryIoCompletion
804e28a0	80573c9e nt!NtQueryKey
804e28a4	8064f3a3 nt!NtQueryMultipleValueKey
804e28a8	8064ac36 nt!NtQueryMutant
804e28ac	8058292c nt!NtQueryObject

Command	
804e28b0	8064f5ad nt!NtQueryOpenSubKeys
804e28b4	80567672 nt!NtQueryPerformanceCounter
804e28b8	806179a9 nt!NtQueryQuotaInformationFile
804e28bc	80580d06 nt!NtQuerySection
804e28c0	80596aac nt!NtQuerySecurityObject
804e28c4	8064942f nt!NtQuerySemaphore
804e28c8	8057afcc nt!NtQuerySymbolicLinkObject
804e28cc	80649b8b nt!NtQuerySystemEnvironmentValue
804e28d0	80649b50 nt!NtSetSystemEnvironmentValueEx
804e28d4	8057f473 nt!NtQuerySystemInformation
804e28d8	8058dc42 nt!NtQuerySystemTime
804e28dc	8058e90d nt!NtQueryTimer
804e28e0	80587979 nt!NtQueryTimerResolution
804e28e4	8056a331 nt!NtQueryValueKey
804e28e8	80571418 nt!NtQueryVirtualMemory
804e28ec	8056fbc3 nt!NtQueryVolumeInformationFile
804e28f0	8058e352 nt!NtQueueApcThread
804e28f4	804e1fb2 nt!NtRaiseException
804e28f8	8064916b nt!NtRaiseHardError
804e28fc	8057044f nt!NtReadFile
804e2900	805da060 nt!NtReadFileScatter
804e2904	8058b527 nt!NtReadRequestData
804e2908	805814b2 nt!NtReadVirtualMemory
804e290c	80579072 nt!NtRegisterThreadTerminatePort
804e2910	805665fb nt!NtReleaseMutant
804e2914	8058ba06 nt!NtReleaseSemaphore
804e2918	80567219 nt!NtRemoveIoCompletion
804e291c	8065c346 nt!NtRemoveProcessDebug
804e2920	8064fa22 nt!NtRenameKey
804e2924	8065017e nt!NtReplaceKey
804e2928	80580514 nt!NtReplyPort
804e292c	8056bd3c nt!NtReplyWaitReceivePort
804e2930	8056b854 nt!NtReplyWaitReceivePortEx

Command	
804e2934	806239f6 nt!NtReplyWaitReplyPort
804e2938	8062c807 nt!NtRequestDeviceWakeup
804e293c	80596862 nt!NtRequestPort
804e2940	8056dd9e nt!NtRequestWaitReplyPort
804e2944	8062c600 nt!NtRequestWakeupLatency
804e2948	805dd720 nt!NtResetEvent
804e294c	8053bd02 nt!NtResetWriteWatch
804e2950	8064fd15 nt!NtRestoreKey
804e2954	806300c0 nt!NtResumeProcess
804e2958	80578f98 nt!NtResumeThread
804e295c	8064fe16 nt!NtSaveKey
804e2960	8064ff01 nt!NtSaveKeyEx
804e2964	8065002e nt!NtSaveMergedKeys
804e2968	80579c0a nt!NtSecureConnectPort
804e296c	80649b8b nt!NtSetBootEntryOrder
804e2970	80649b8b nt!NtSetBootEntryOrder
804e2974	8062e94f nt!NtSetContextThread
804e2978	8065de8e nt!NtSetDebugFilterState
804e297c	805b21fb nt!NtSetDefaultHardErrorPort
804e2980	805ad7b7 nt!NtSetDefaultLocale
804e2984	805ad75e nt!NtSetDefaultUILanguage
804e2988	80617635 nt!NtSetEaFile
804e298c	806699fd nt!NtSetEvent
804e2990	8056ca46 nt!NtSetEventBoostPriority
804e2994	8064a5c1 nt!NtSetHighEventPair
804e2998	8064a4e5 nt!NtSetHighWaitLowEventPair
804e299c	8065bce7 nt!NtSetInformationDebugObject
804e29a0	8057e4da nt!NtSetInformationFile
804e29a4	805a01c4 nt!NtSetInformationJobObject
804e29a8	8064ef06 nt!NtSetInformationKey
804e29ac	80581ba6 nt!NtSetInformationObject
804e29b0	80570e2d nt!NtSetInformationProcess
804e29b4	8056c62e nt!NtSetInformationThread
804e29b8	80570e2d nt!NtSetInformationThread

```

Command X
804e29b8 805a75bc nt!NtSetInformationToken
804e29bc 8064a7db nt!NtSetIntervalProfile
804e29c0 8056c4e3 nt!NtSetIoCompletion
804e29c4 8062f1a3 nt!NtSetIdEntries
804e29c8 8064a357 nt!NtSetLowEventPair
804e29cc 8064a473 nt!NtSetLowHighEventPair
804e29d0 80617981 nt!NtSetQuotaInformationFile
804e29d4 80598227 nt!NtSetSecurityObject
804e29d8 80649e50 nt!NtSetSystemEnvironmentValue
804e29dc 80649b50 nt!NtSetSystemEnvironmentValueEx
804e29e0 805a6aa9 nt!NtSetSystemInformation
804e29e4 8065078b nt!NtSetSystemPowerState
804e29e8 80648ab3 nt!NtSetSystemTime
804e29ec 805e02da nt!NtSetThreadExecutionState
804e29f0 804e570b nt!NtSetTimer
804e29f4 805e0960 nt!NtSetTimerResolution
804e29f8 805a98db nt!NtSetUuidSeed
804e29fc 8057daf3 nt!NtSetValueKey
804e2a00 80617ec3 nt!NtSetVolumeInformationFile
804e2a04 806481ff nt!NtShutdownSystem
804e2a08 80517369 nt!NtSignalAndWaitForSingleObject
804e2a0c 8064aa44 nt!NtStartProfile
804e2a10 8064abfd nt!NtStopProfile
804e2a14 80630065 nt!NtSuspendProcess
804e2a18 805e0506 nt!NtSuspendThread
804e2a1c 8064a45d nt!NtSystemDebugControl
804e2a20 80630999 nt!NtTerminateJobObject
804e2a24 80585851 nt!NtTerminateProcess
804e2a28 80578037 nt!NtTerminateThread
804e2a2c 80578a7f nt!NtTestAlert
804e2a30 80545d00 nt!NtTraceEvent
804e2a34 80649b9f nt!NtTranslateFilePath
804e2a38 8061a306 nt!NtUnloadDriver
804e2a3c 8061a306 nt!NtUnloadDriver
kd>

```

```

Command X
804e2a3c 8064ea6a nt!NtUnloadKey
804e2a40 8064ec9b nt!NtUnloadKeyEx
804e2a44 80590abc nt!NtUnlockFile
804e2a48 80627907 nt!NtUnlockVirtualMemory
804e2a4c 8057cd86 nt!NtUnmapViewOfSection
804e2a50 805be9a8 nt!NtVdmControl
804e2a54 8065ba30 nt!NtWaitForDebugEvent
804e2a58 80566846 nt!NtWaitForMultipleObjects
804e2a5c 805662fc nt!NtWaitForSingleObject
804e2a60 8064a409 nt!NtWaitHighEventPair
804e2a64 8064a39f nt!NtWaitLowEventPair
804e2a68 80570785 nt!NtWriteFile
804e2a6c 805d9c96 nt!NtWriteFileGather
804e2a70 8058b714 nt!NtWriteRequestData
804e2a74 805815aa nt!NtWriteVirtualMemory
804e2a78 804f5142 nt!NtYieldExecution
804e2a7c 805cbc2d nt!NtCreateKeyedEvent
804e2a80 80584ab9 nt!NtOpenKeyedEvent
804e2a84 8064b1d1 nt!NtReleaseKeyedEvent
804e2a88 8064b46c nt!NtWaitForKeyedEvent
804e2a8c 8062dbf1 nt!NtQueryPortInformationProcess
804e2a90 57565355
804e2a94 f1241d8b
804e2a98 848dfff0
804e2a9c ffd8002a
804e2aa0 1c433bff
804e2aa4 e8541673
804e2aa8 0000f4b7
804e2aac 850fc00b
804e2ab0 000000d8
804e2ab4 a31c418b
804e2ab8 ffdff008
804e2abc 012cb3ff
804e2ab4 00000000
804e2ac0 938b0000
804e2ac4 00000134

```

Q3:

Yes, the debugger is capable of identifying these hooked functions, in this case the attack hooked “nt!NtDeleteValueKey”, “nt!NtLoadDriver”, “nt!NtOpenProcess”, and “nt!NtTerminateProcess”.

Command	X
804e276c	80566590 nt!NtDelayExecution
804e2710	8058bef5 nt!NtDeleteAtom
804e2714	80649b77 nt!NtDeleteBootEntry
804e2718	805d610b nt!NtDeleteFile
804e271c	80595aba nt!NtDeleteKey
804e2720	8063b1b7 nt!NtDeleteObjectAuditAlarm
804e2724	faedf58Unable to load image SSDHook.sys, Win32 error 0n2
SSDHook.dll	805936da nt!NtDeleteValueKey
804e2728	805796db nt!NtDeviceIoControlFile
804e272c	805c5e6b nt!NtDisplayString
804e2730	805749da nt!NtDuplicateObject
804e2734	8058081b nt!NtDuplicateToken
804e2738	806498b8 nt!NtSetBootEntryOrder
804e273c	80573f95 nt!NtEnumerateKey
804e2740	80649b63 nt!NtEnumerateSystemEnvironmentValuesEx
804e2744	805819c3 nt!NtEnumerateValueKey
804e2748	80625e50 nt!NtExtendSection
804e274c	806afad0 nt!NtFilterToken
804e2750	805961e0 nt!NtFindAtom
804e2754	8058c894 nt!NtFlushBuffersFile
804e2758	8056e7aa nt!NtFlushInstructionCache
804e275c	8059d120 nt!NtFlushKey
804e2760	805960e1 nt!NtFlushVirtualMemory
804e2764	80627893 nt!NtFlushWriteBuffer
804e2768	806273e4 nt!NtFreeUserPhysicalPages
804e276c	80569c2d nt!NtFreeVirtualMemory
804e2770	805771f8 nt!NtFsControlFile
804e2774	805e056b nt!NtGetContextThread
804e2778	8062c893 nt!NtGetDevicePowerState
804e277c	8059f05c nt!NtGetDevicePowerState
804e2780	8053b86d nt!NtGetEventWatch
804e2784	805dc22e nt!NtImpersonateAnonymousToken
804e2788	8058b1e2 nt!NtImpersonateClientOfPort
804e278c	805817c1 nt!NtImpersonateThread
kds	804e2790
	80565990 nt!NtDelayExecution
	804e2710
	8058bef5 nt!NtDeleteAtom
	804e2714
	80649b77 nt!NtDeleteBootEntry
	804e2718
	805d610b nt!NtDeleteFile
	804e271c
	80595aba nt!NtDeleteKey
	804e2720
	8063b1b7 nt!NtDeleteObjectAuditAlarm
	804e2724
	805936da nt!NtDeleteValueKey
	804e2728
	805796db nt!NtDeviceIoControlFile
	804e272c
	805c5e6b nt!NtDisplayString
	804e2730
	805749da nt!NtDuplicateObject
	804e2734
	8058081b nt!NtDuplicateToken
	804e2738
	806498b8 nt!NtSetBootEntryOrder
	804e273c
	80573f95 nt!NtEnumerateKey
	804e2740
	80649b63 nt!NtEnumerateSystemEnvironmentValuesEx
	804e2744
	805819c3 nt!NtEnumerateValueKey
	804e2748
	80625e50 nt!NtExtendSection
	804e274c
	806afad0 nt!NtFilterToken
	804e2750
	805961e0 nt!NtFindAtom
	804e2754
	8058c894 nt!NtFlushBuffersFile
	804e2758
	8056e7aa nt!NtFlushInstructionCache
	804e275c
	8059d120 nt!NtFlushKey
	804e2760
	805960e1 nt!NtFlushVirtualMemory
	804e2764
	80627893 nt!NtFlushWriteBuffer
	804e2768
	806273e4 nt!NtFreeUserPhysicalPages
	804e276c
	80569c2d nt!NtFreeVirtualMemory
	804e2770
	805771f8 nt!NtFsControlFile
	804e2774
	805e056b nt!NtGetContextThread
	804e2778
	8062c893 nt!NtGetDevicePowerState
	804e277c
	8059f05c nt!NtGetDevicePowerState
	804e2780
	8053b86d nt!NtGetEventWatch
	804e2784
	805dc22e nt!NtImpersonateAnonymousToken
	804e2788
	8058b1e2 nt!NtImpersonateClientOfPort
	804e278c
	805817c1 nt!NtImpersonateThread
	804e2790
	8056f30a nt!NtInitializeRegistry

Command X	804e2775 8052c0f6 nt!NtListenPort	804e279c 8062c87a nt!NtIsSystemResumeAutomatic
804e277a0 805a95b1 nt!NtListenPort	804e27a0 805a95b1 nt!NtListenPort	804e27a0 805a95b1 nt!NtListenPort
804e277a4 faebd08c SSOTHook+0x108c	804e27a4 805a29bd nt!NtLoadDriver	804e27a8 805adcb8 nt!NtLoadKey
804e277a8 805adcb8 nt!NtLoadKey	804e27ac 805adaf8 nt!NtLoadKey2	804e27b0 8059095c nt!NtLockFile
804e277ac 805adaf8 nt!NtLoadKey2	804e27b4 805afca0 nt!NtLockProductActivationKeys	804e27b8 805d0d8f nt!NtLockRegistryKey
804e277b0 8059095c nt!NtLockFile	804e27bc 805af122 nt!NtLockVirtualMemory	804e27c0 8059eb3 nt!NtMakePermanentObject
804e277b4 805afca0 nt!NtLockProductActivationKeys	804e27c4 8059eb60 nt!NtMakeTemporaryObject	804e27c8 8062651b nt!NtMapUserPhysicalPages
804e277b8 805d0d8f nt!NtLockRegistryKey	804e27cc 806269ef nt!NtMapUserPhysicalPagesScatter	804e27d0 8057cb31 nt!NtMapViewOfSection
804e277bc 805af122 nt!NtLockVirtualMemory	804e27d4 80649b77 nt!NtDeleteBootEntry	804e27d8 80591d22 nt!NtNotifyChangeDirectoryFile
804e277c0 8059eb3 nt!NtMakePermanentObject	804e27dc 80591a6b nt!NtNotifyChangeKey	804e27e0 80591b34 nt!NtNotifyChangeMultipleKeys
804e277c4 8059eb60 nt!NtMakeTemporaryObject	804e27e4 8057b28f nt!NtOpenDirectoryObject	804e27e8 80581b30 nt!NtOpenEvent
804e277c8 8062651b nt!NtMapUserPhysicalPages	804e27ec 8064a2cd nt!NtOpenEventPair	804e27f0 8056f91b nt!NtOpenFile
804e277cc 806269ef nt!NtMapUserPhysicalPagesScatter	804e27f4 80616e9f nt!NtOpenIoCompletion	804e27f8 80630823 nt!NtOpenJobObject
804e27d0 8057cb31 nt!NtMapViewOfSection	804e27fc 80569080 nt!NtOpenKey	804e2800 8057778e nt!NtOpenMutant
804e27d4 80649b77 nt!NtDeleteBootEntry	804e2804 80595ba5 nt!NtOpenObjectAuditAlarm	804e2808 80574bc1 nt!NtOpenProcess
804e27d8 80591d22 nt!NtNotifyChangeDirectoryFile	804e280c 80571121 nt!NtOpenProcessToken	804e2810 8057131a nt!NtOpenProcessTokenEx
804e27dc 80591a6b nt!NtNotifyChangeKey	804e2814 8056e583 nt!NtOpenSection	804e2818 805daca3 nt!NtOpenSemaphore
804e27e0 80591b34 nt!NtNotifyChangeMultipleKeys	804e281c 8057b15b nt!NtOpenSymbolicLinkObject	804e2820 80590cfc nt!NtOpenThread
804e27e4 8057b28f nt!NtOpenDirectoryObject	804e2824 80570bbe nt!NtOpenThreadToken	804e2828 80570b2f nt!NtOpenThreadTokenEx
804e27e8 80581b30 nt!NtOpenEvent	804e282c 8064a103 nt!NtOpenTimer	804e2830 8059be7a nt!NtPlugPlayControl
804e27ec 8064a2cd nt!NtOpenEventPair	804e2834 8059abdc nt!NtPowerInformation	804e2838 80596c0c nt!NtPrivilegeCheck
804e27f0 8056f91b nt!NtOpenFile	804e283c 8059dcd8 nt!NtPrivilegeObjectAuditAlarm	804e2840 805a96f4 nt!NtPrivilegedServiceAuditAlarm
804e27f4 80616e9f nt!NtOpenIoCompletion	804e2844 80574f70 nt!NtProtectVirtualMemory	804e2848 805d9a2e nt!NtPulseEvent
804e27f8 80630823 nt!NtOpenJobObject	804e284c 805707ea nt!NtQueryAttributesFile	804e2850 80649b8b nt!NtSetBootEntryOrder
804e27fc 80569080 nt!NtOpenKey	804e2854 80649b8b nt!NtSetBootEntryOrder	804e2858 804f97c5 nt!NtQueryDebugFilterState
804e2800 8057778e nt!NtOpenMutant	804e285c 80566d02 nt!NtQueryDefaultLocale	804e2860 80581f1f nt!NtQueryDefaultUILanguage
804e2804 80595ba5 nt!NtOpenObjectAuditAlarm	804e2864 8057d37b nt!NtQueryDirectoryFile	804e2868 80587913 nt!NtQueryDirectoryObject
804e2808 80574bc1 nt!NtOpenProcess	804e286c 805770e8 nt!NtQueryEvent	804e2870 8058004a nt!NtQueryFullAttributesFile
804e280c 80571121 nt!NtOpenProcessToken	804e2874 805d57e8 nt!NtQueryInformationAtom	804e2878 8057bc38 nt!NtQueryInformationFile
804e2810 8057131a nt!NtOpenProcessTokenEx	804e287c 80583d25 nt!NtQueryInformationJobObject	804e2880 80623917 nt!NtQueryInformationPort
804e2814 8056e583 nt!NtOpenSection	804e2884 80623917 nt!NtQueryInformationProcess	
804e2818 805daca3 nt!NtOpenSemaphore		
804e281c 8057b15b nt!NtOpenSymbolicLinkObject		
804e2820 80590cfc nt!NtOpenThread		
804e2824 80570bbe nt!NtOpenThreadToken		
804e2828 80570b2f nt!NtOpenThreadTokenEx		
804e282c 8064a103 nt!NtOpenTimer		
804e2830 8059be7a nt!NtPlugPlayControl		
804e2834 8059abdc nt!NtPowerInformation		
804e2838 80596c0c nt!NtPrivilegeCheck		
804e283c 8059dcd8 nt!NtPrivilegeObjectAuditAlarm		
804e2840 805a96f4 nt!NtPrivilegedServiceAuditAlarm		
804e2844 80574f70 nt!NtProtectVirtualMemory		
804e2848 805d9a2e nt!NtPulseEvent		
804e284c 805707ea nt!NtQueryAttributesFile		
804e2850 80649b8b nt!NtSetBootEntryOrder		
804e2854 80649b8b nt!NtSetBootEntryOrder		
804e2858 804f97c5 nt!NtQueryDebugFilterState		
804e285c 80566d02 nt!NtQueryDefaultLocale		
804e2860 80581f1f nt!NtQueryDefaultUILanguage		
804e2864 8057d37b nt!NtQueryDirectoryFile		
804e2868 80587913 nt!NtQueryDirectoryObject		
804e286c 805770e8 nt!NtQueryEvent		
804e2870 8058004a nt!NtQueryFullAttributesFile		
804e2874 805d57e8 nt!NtQueryInformationAtom		
804e2878 8057bc38 nt!NtQueryInformationFile		
804e287c 80583d25 nt!NtQueryInformationJobObject		
804e2880 80623917 nt!NtQueryInformationPort		
804e2884 80623917 nt!NtQueryInformationProcess		
kd>		
Locals	Ln 286, Col 19 12,051 characters Plain text 100% Windows (CRLF) UTF-8	
Command X	804e2a20 80630999 nt!NtTerminateJobObject	804e2a20 80630999 nt!NtTerminateJobObject
804e2a20 80630999 nt!NtTerminateJobObject	804e2a24 80578037 nt!NtTestAlert	804e2a28 80545d00 nt!NtTraceEvent
804e2a24 faebd006 SSOTHook+0x1006	804e2a2c 80649b9f nt!NtTranslateFilePath	804e2a30 8061a306 nt!NtUnloadDriver
804e2a28 80578037 nt!NtTestAlert	804e2a34 8064e306 nt!NtUnloadKey	804e2a38 8064ec9b nt!NtUnloadKeyEx
804e2a2c 80578a7f nt!NtTestAlert	804e2a3c 8064e306 nt!NtUnloadKey	804e2a40 80590abc nt!NtUnlockFile
804e2a30 80545d00 nt!NtTraceEvent	804e2a44 80590abc nt!NtUnlockFile	804e2a48 80627907 nt!NtUnlockVirtualMemory
804e2a34 80649b9f nt!NtTranslateFilePath	804e2a4c 8057c6b6 nt!NtUnmapViewOfSection	804e2a50 805b9a8 nt!NtVdmControl
804e2a38 8061a306 nt!NtUnloadDriver	804e2a54 805b9a8 nt!NtVdmControl	804e2a58 80566846 nt!NtWaitForMultipleObjects
804e2a3c 8064e306 nt!NtUnloadKey	804e2a5c 805662fc nt!NtWaitForSingleObject	804e2a60 8064a409 nt!NtWaitHighEventPair
804e2a40 8064ec9b nt!NtUnloadKeyEx	804e2a64 8064a39f nt!NtWaitLowEventPair	804e2a68 8057e785 nt!NtWriteFile
804e2a44 80590abc nt!NtUnlockFile	804e2a6c 805d9c96 nt!NtWriteFileGather	804e2a70 8058b714 nt!NtWriteRequestData
804e2a48 80627907 nt!NtUnlockVirtualMemory	804e2a74 805815aa nt!NtWriteVirtualMemory	804e2a78 804f5142 nt!NtYieldExecution
804e2a4c 8057c6b6 nt!NtUnmapViewOfSection	804e2a7c 805bcb2d nt!NtCreateKeyedEvent	804e2a80 80584ab0 nt!NtOpenKeyedEvent
804e2a50 805b9a8 nt!NtVdmControl	804e2a84 8064b1d1 nt!NtReleaseKeyedEvent	804e2a88 8064b46c nt!NtWaitForKeyedEvent
804e2a54 805b9a8 nt!NtVdmControl	804e2a8c 8062dbf1 nt!NtQueryPortInformationProcess	804e2a90 57565355
804e2a58 80566846 nt!NtWaitForMultipleObjects	804e2a94 f1241d8b	804e2a98 848dffdf
804e2a5c 805662fc nt!NtWaitForSingleObject	804e2a9c ffd00024	804e2aa0 1c433bff
804e2a60 8064a409 nt!NtWaitHighEventPair	804e2aa4 e8541673	
804e2a64 8064a39f nt!NtWaitLowEventPair		
804e2a68 8057e785 nt!NtWriteFile		
804e2a6c 805d9c96 nt!NtWriteFileGather		
804e2a70 8058b714 nt!NtWriteRequestData		
804e2a74 805815aa nt!NtWriteVirtualMemory		
804e2a78 804f5142 nt!NtYieldExecution		
804e2a7c 805bcb2d nt!NtCreateKeyedEvent		
804e2a80 80584ab0 nt!NtOpenKeyedEvent		
804e2a84 8064b1d1 nt!NtReleaseKeyedEvent		
804e2a88 8064b46c nt!NtWaitForKeyedEvent		
804e2a8c 8062dbf1 nt!NtQueryPortInformationProcess		
804e2a90 57565355		
804e2a94 f1241d8b		
804e2a98 848dffdf		
804e2a9c ffd00024		
804e2aa0 1c433bff		
kd>		

Q4:

Based on the four hooked functions we can get an idea of what the rootkit can or cannot do. It gains some sort of control during the opening of processes so it can do things such as

prevent the opening of a process, read what processes are being opened, or redirect the opening elsewhere. It also has a hook into the termination which makes it much harder to be removed from a system since it can block its termination. In tandem with opening if it could open bloatware or other viruses it could protect them by blocking termination of these processes. It hooks into the loading of drivers so it can redirect to using its own drivers if it has any which would allow the rootkit the high-level privileges as “root” or block out the anti-virus or other system drivers for malicious purposes. Finally, the deletion of value keys gives it an additional layer of protection from the user by denying deletion of its hooked registers. With these four functions, the rootkit has essentially total control over the system.